

CS527-Computer System.

Lab5

25th Aug 2025

Points: 50. Total time: 120 minutes

Team size: 1 (to be done individually)

Submission instructions: Submit a zip file(file name should be like: entry_no.zip), that contains all the source code files, a Makefile. Compilation and execution instructions can be given in a text file called "readme".

Instructions:

- No LLM/google search will be allowed.
- You can use only <https://cppreference.com/> website for reference
- You can use "your" previous lab codes (lab1, Lab2, Lab3, Lab4).
- No mobile phone allowed.

Description:

- The communication protocol is the same as lab 4

Modified system setup is:

- There are four devices: Keyboard, display, alarm and push button. Details of these devices are given below.
 - a. Keyboard: memory address starts at 0xFFFF0000. Total memory is 128 bytes. Only read requests can be sent to the keyboard (by sending a read request to 0xFFFF0000). After receiving the read request, the keyboard sends a string of characters terminated by a null character. The keyboard can call a scanf to get a string from the user.
 - b. Display: memory address starts at 0xFFFF0100. Total memory is 512 bytes. Only write requests can be sent to the display (by sending a write request to any of the following addresses: 0xFFFF0100 to 0xFFFF02FF). When a request is received, this device uses standard printf function to print the information as string
 - c. Alarm: Memory address starts at 0xFFFF0300. Total memory is one byte. Only write requests can be sent. When a write request is received, this device takes it like an unsigned integer, and prints "Alarm\n" number of times mentioned in the byte received. (In the real scenario, it shows the duration of time the alarm will ring).
 - d. Push button: Memory address starts at 0xFFFF0301. Total memory is one byte. Only read requests can be sent. When a read request is received, this device sends 0 or 1 randomly.
- Modified application (host side) interface for drivers. Application has only two functions to interact with lower layers of protocol: read and write, with the following signature:
 - a. void read(unsigned int address, char *message, unsigned int size); // function expects data will be received in message string.

- b. `void write(unsigned int address, char *message, unsigned int size);`
- Following is the pseudo code of application to be written
 - a. Sends "Welcome to CS527: Computer Systems \n Let's have fun\n" to display
 - b. Send "Write a random number" to display.
 - c. Expect a number of the user by sending a read request at the keyboard.
 - d. If number received is less than 24, the a read request from push button is sent
 - i. If push button sends 1, write request to alarm is sent with number of received
 - ii. Else go to 'b'
 - e. Else go to 'b'

Hints:

- The Bus will work as an address decoder.
- Since information is taken/input/stored as characters, it may be required to convert to integers, based on requirement